

22 Inducible CRISPR-based Genome Editing for the Characterization of Cancer Genes

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22.1 Introduction

Recent advances in sequencing technology have enabled us to study the molecular alterations in tumors to an unprecedented degree of granularity. At the same time, CRISPR/Cas9 systems have proven to be extraordinary tools that may be used to generate genetic models in animal and cell culture systems that improve our knowledge of tumor formation, development and progression. Importantly, these systems have also facilitated the identification of genes required for the survival of cancer cells. Compared to other gene editing tools, CRISPR/Cas9 methods are easy to implement and allow for highly efficient and specific gene editing. These superior features of CRISPR/Cas9 methods have spurred the exceptional development of this technology and its widespread adoption. Ultimately, CRISPR/Cas9 methods have played a critical role in narrowing the gap between the genomic and functional characterization of tumors.

In general, CRISPR/Cas9 methods rely on a *Streptococcus pyogenes* Type II CRISPR system in which the central components are a guide RNA and a Cas9 enzyme (Doudna and Charpentier, 2014; Doudna and Sontheimer, 2014; Jinek *et al.*, 2012; Wright *et al.*, 2016). In order for the Cas9 to be activated, the single-guide RNA (sgRNA) has to be fused to two key RNA units: the CRISPR RNA and the transactivating CrRNA (Doudna and Charpentier, 2014; Doudna and Sontheimer, 2014; Jinek *et al.*, 2012; Wright *et al.*, 2016). When introduced to target cells, the sgRNA promptly recognizes complementary sequences of DNA, unwinds the double-stranded DNA and enables its cleavage by Cas9 (Doudna and Charpentier, 2014; Doudna and Sontheimer, 2014; Jinek *et al.*, 2012; Wright *et al.*, 2016). Consequently, the generation of double-stranded DNA breaks triggers error-prone non-homologous end joining (NHEJ) repair or, in some cases, homology-directed repair (HDR), leading to the formation of indels in the target DNA (Doudna and Charpentier, 2014; Doudna and Sontheimer, 2014; Jinek *et al.*, 2012; Wright *et al.*, 2016).

Cancer mutations can result from single nucleotide variations, small insertions or deletions, large deletions and structural alterations (e.g. chromosomal copy

number alterations, translocations and inversions). To model these diverse genetic aberrations, novel CRISPR/Cas9 applications have been developed and adapted with different degrees of success (Wright *et al.*, 2016). In addition, although CRISPR/Cas9 methods have been traditionally used to generate stable genetic alterations, the integration of a mutated, enzymatically dead Cas9 has enabled the repression or activation of the expression of genes of interest (Chavez *et al.*, 2015; Konermann *et al.*, 2015). This is a particularly interesting implementation of the gene editing tool, as it allows the reversible expression of genes of interest.

Since 2013 when CRISPR/Cas9 was first reported as a tool to induce gene editing, a plethora of studies have verified the efficient and robust employment of this technology in cancer biology (Doudna and Charpentier, 2014; Sternberg *et al.*, 2014; Wang *et al.*, 2015). The CRISPR/Cas9 system has mostly been utilized for loss-of-function studies that facilitate the identification of novel tumor-suppressor genes and genes that are essential for tumor cells' survival (Doudna and Charpentier, 2014; Sternberg *et al.*, 2014; Wang *et al.*, 2015).

In addition to *in vitro* studies, CRISPR/Cas9 methods have also been used to generate genetic mouse models of both inherited germline mutations and somatic mutations (Singh *et al.*, 2015; Wright *et al.*, 2016; Yang *et al.*, 2014). The creation of genetically engineered mouse models (GEMMs) via the manipulation of embryonic stem (ES) cells has become the standard for generating cancer models over the last three decades and has been easily adapted to the use of CRISPR/Cas9 methods (Singh *et al.*, 2015; Wright *et al.*, 2016; Yang *et al.*, 2014). Although very powerful, one of the major drawbacks of traditional gene targeting has been the low efficiency of double-strand breaks (DSB) at targeting sites. CRISPR/Cas9 circumvents this limitation by increasing the frequency of DSB at specific loci. In 2013, Yang *et al.* (2013) highlighted the advantages of this GEMM generation technique by showing the feasibility of concurrently disrupting five genes in ES cells using a CRISPR/Cas9-based method. Furthermore, the use of tissue-specific and inducible CRISPR/Cas9 systems has also enabled the development of highly sophisticated models based on tissue-specific and temporal genetic manipulations of genes of interest, as well as models based on the constitutive all-body expression/deletion of genes of interest (Dow *et al.*, 2015; Utomo *et al.*, 1999).

More recently, it has been demonstrated that CRISPR/Cas9 could be used to introduce gene editing directly into zygotes at pronuclear states with an efficiency of over 80% (Yang *et al.*, 2013; 2014). This has been a major breakthrough, as this approach doesn't require the cumbersome manipulation of ES cells and facilitates the more rapid generation of animal models. Moreover, while the generation of GEMMs using traditional gene-targeting methods in ES cells requires several rounds of breeding over the span of months, the CRISPR/Cas9-based approach allows the generation of mice with the desired genotype in just a few weeks. Another advantage of this approach is the possibility of creating GEMMs with animals that formerly could not be used for genetically modified modeling, including pigs, rats and non-human primates.

Although faster and more affordable, the traditional CRISPR/Cas9 system mostly relies on NHEJ repair mechanisms. This dependence impacts the nature of the mutations that can be generated, as NHEJ-based gene disruption can carry different indels. Consequently, different mutations can be introduced to the two alleles of a target gene, and mice in the same litter can harbor very different mutations. To overcome this problem, HDR-based CRISPR/Cas9 methods have been developed in which single-stranded DNA templates with sgRNA and Cas9 mRNA have been co-injected into zygotes to introduce a defined point mutation (Yang *et al.*, 2014). Although this process is less efficient compared to NHEJ-based CRISPR/Cas9 methods, it allows targeted mutation insertions to be achieved with 10–50% efficiency depending on the size of the fragments that are being inserted (Yang *et al.*, 2014).

In addition to the aforementioned applications of CRISPR/Cas9, multiple groups have developed efficient methods for using CRISPR/Cas9 for somatic gene editing *in vivo*. In one such study, Bell *et al.* leveraged the unique capability of liver cells to uptake naked DNA and demonstrated the possibility of transducing Cas9 and sgRNAs in liver cells to induce indel formation using hydrodynamic tail injection methods (Bell *et al.*, 2007). As this particular method is practically liver-specific, alternative techniques have been developed to enable gene editing in other organs based on viral delivery. These techniques involve the construction of adeno-associated-based vectors (AAV), as well as lentiviral vectors (Swiech *et al.*, 2015). As the size of Cas9 can limit the packaging of the viral particles, several groups have now generated mice expressing Cas9, either constitutively or conditionally, using lox-stop-lox strategies or using doxycycline-inducible promoters (Dow *et al.*, 2015, Utomo *et al.*, 1999). This approach has been shown to be more efficient as it enables the viral-mediated transduction of only the sgRNAs.

Recent technical advances have expanded the use of CRISPR/Cas9 for the generation of mutations other than indels (Platt *et al.*, 2014). The Jacks Lab, for example, have shown that KrasG12D mutations could be generated in the lungs of mice upon adenoviral delivery of sgRNA, although only with an efficiency of 0.1% (DuPage *et al.*, 2009). Similar results were obtained in the livers of mice upon hydrodynamic delivery (Xue *et al.*, 2014). For the introduction of larger deletions, the Ventura Lab was able to generate deletions of up to 1.5 MB (Maddalo *et al.*, 2014). Specifically, they successfully introduced concomitant double-strand DNA breaks at intron 14 of *Eml4* (corresponds to intron 13 of *EML4*) and at intron 19 of *Alk* using a CRISPR/Cas9 system in order to model the most common *EML4*–*ALK* variant in human non-small-cell lung cancers (NSCLC; Xue *et al.*, 2014).

As an alternative strategy to model genetic mutations *in vivo*, mosaic models in which isolated cells can be genetically modified *in vitro* and then transplanted into recipient mice have been developed (Dow *et al.*, 2012). These methods are notable because, much like siRNA-based systems, they are amenable for unbiased multiplex large-scale genomic screening.

Tumors harbor diverse genetic aberrations that accumulate during the process of tumorigenesis (Hanahan and Weinberg, 2011). While some of these mutations accumulate during clonal expansions, others contribute to the evolution and

establishment of particular clonal populations (Hanahan and Weinberg, 2011). Therefore, a major challenge in tumor biology is elucidating the contributions of each of these cancer mutations during cancer initiation and progression. Moreover, several of these mutations have become essential for cancer cell survival, rendering the characterization of these mutations difficult – especially *in vivo*. Nevertheless, the study of essential genes in cancer is particularly salient as it could lead to the identification of novel therapeutic targets.

To overcome these hurdles, CRISPR/Cas9-inducible systems have been developed (Davis *et al.*, 2015; Dow *et al.*, 2015; Nihongaki *et al.*, 2015; Wright *et al.*, 2015). These systems are based on the transcriptional regulation of Cas9, for example, through the use of a DOX-induced sgRNA or on post-transcriptional methods. In the latter case multiple strategies have been employed including: (1) the generation of a split Cas9 protein that can be restored and activated in the presence of rapamycin, (2) the insertion of a 4-hydroxytamoxifen (4-HT)-dependent intein that can be spliced-out upon treatment with tamoxifen or (3) by fusing the light-inducible heterodimerizing proteins CRY2 and CIB1 to Cas9 or a mutated version of Cas9 (Davis *et al.*, 2015; Dow *et al.*, 2015; Nihongaki *et al.*, 2015; Wright *et al.*, 2015).

In addition, methods based on the temporal control of CRISPR/Cas9 activity by conditional destabilization of Cas9 have proved promising (Senturk *et al.*, 2017). The first conditional destabilization method of proteins based on the destabilization domain (DD)-based technology stemmed from the identification of specific destabilizing mutations in the human FKBP12 protein, and the concurrent development of a family of highly specific synthetic FKBP12 ligands that are cell permeable and non-toxic in cultured cells and animals (Banaszynski *et al.*, 2006; 2008). The small-molecule ligand Shield-1 was shown to bind mutant FKBP12 more tightly than the wild-type by three orders of magnitude, and to prevent proteasome-induced degradation of DD-GFP and other fusion proteins (Bonger *et al.*, 2011; Clackson *et al.*, 1998). This DD system has been demonstrated to work well for a variety of proteins, including kinases, cell cycle regulatory proteins and small GTPases expressed in cultured cells and mice (Banaszynski *et al.*, 2006; 2008).

By fusing the FKBP12-derived destabilizing domain to Cas9 (DD-Cas9), Senturk *et al.* demonstrated that this means of conditionally regulating protein stability could be exploited for rapid and reversible Cas9 expression *in vitro* (Senturk *et al.*, 2017).

Specifically, they showed that fusing an FKBP12-derived destabilizing domain to Cas9 (DD-Cas9) enables conditional Cas9 expression and temporal control of gene editing in the presence of an FKBP12 synthetic ligand (Figure 22.1A; Senturk *et al.*, 2017). This technique could be employed to conditionally target a variety of genes controlling diverse biological processes such as oncogenic transformation (EGFR), DNA damage responses (TP53), mitochondria metabolism (CypD) and DNA replication/repair (RPA3). By targeting the RPA3 gene and EGFR in “EGFR-addicted” cells, they demonstrated the ability of the DD-Cas9 system to identify genes that are essential for sustained cell growth and survival (Senturk *et al.*, 2017).

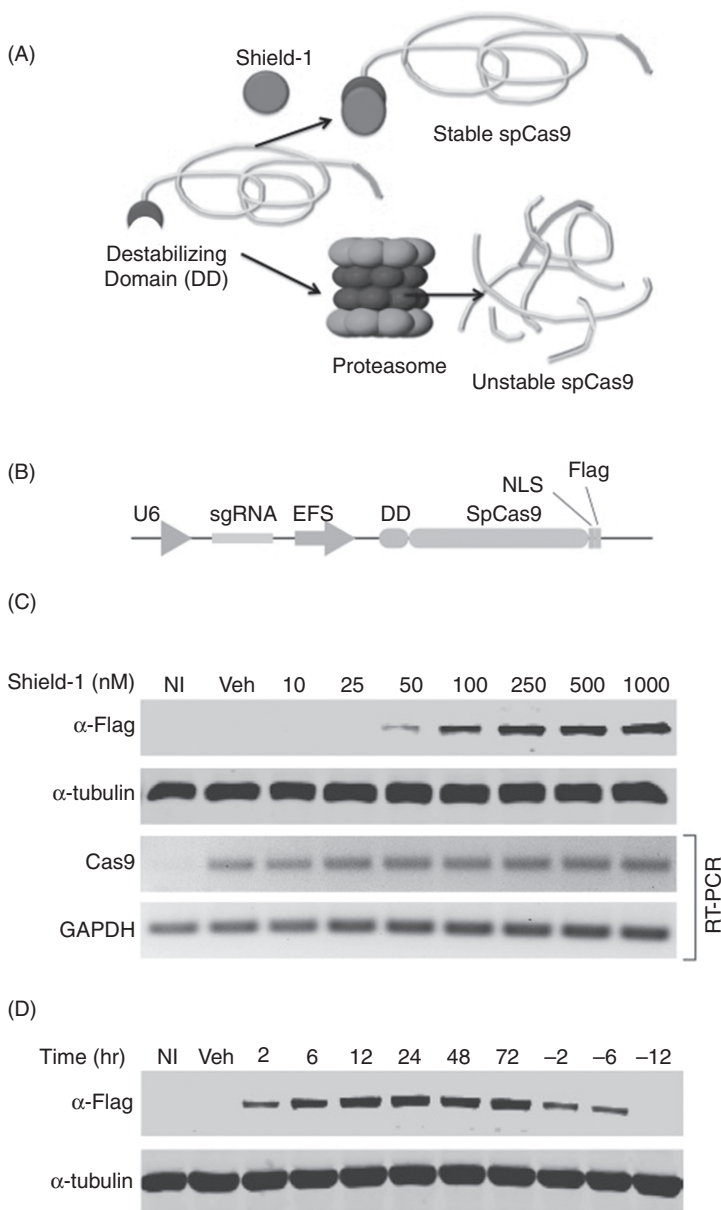


Figure 22.1 Establishment of an inducible lentiviral guide RNA expression system based on conditionally destabilized Cas9. (A) Diagram of conditionally destabilized Cas9 (DD-Cas9). Cas9 is fused to the destabilizing domain of mutant FKBP12 that triggers the rapid degradation of the fused-protein. The synthetic ligand (Shield-1) prevents this destabilizing effect and thus degradation of Cas9. Note, a Flag tag is present at the C-terminal of Cas9. (B) Schematic of the lentiviral construct utilized in our studies. (C) Ligand-dependent stabilization of DD-Cas9. Cells expressing destabilized Cas9 (DD-Cas9) and uninfected cells (NI) were either mock-treated (vehicle) or treated with increasing concentrations of Shield-1, as indicated. To measure the level of expression of Cas9, lysates were analyzed by western blot with an anti-Flag antibody. Various degrees of Shield-1-induced stabilization could be observed in a dose-dependent manner. DD-Cas9 was not detected in lysate from mock-treated cells. To verify that the destabilization of DD-Cas9 occurs post-transcriptionally, DD-Cas9 expression was also verified by RT-PCR analysis using Cas9-specific primers and GAPDH as the loading control. (D) Rapid and reversible regulation of destabilized DD-Cas9. A549 cells transduced with the DD-Cas9 lentiviral vector and uninfected cells were treated with Shield-1 (200 nM) for different time points as indicated. Samples were assayed by western blot analysis at the indicated time points (Senturk *et al.*, 2017). (A black-and-white version of this figure will appear in some formats. For the color version, please refer to the plate section.)

One unique feature of this method is that the Cas9 protein's expression is conditionally regulated independently of its mRNA expression. Hence, this vector system can be easily adapted to co-express – from the same promoter – DD-Cas9 with any other gene of interest, without the latter being co-modulated. Senturk *et al.* demonstrated that this approach could be used for generating traceable systems and for interrogating genes' functional interactions *in vivo* (Senturk *et al.*, 2017). Moreover, when they coupled DD-Cas9 with a conditional Cre allele (Cre-ER^{T2}), they showed this system could be utilized to facilitate the analysis of genes that modulate disease onset and progression in a variety of pre-existing mouse models of human disease based on the Cre-Lox system (Senturk *et al.*, 2017).

22.2 Methodology

Fusing Cas9 to a destabilizing domain provides a highly efficient and easily scalable new modality for the temporal control of gene editing. In principle, this technique is suited for use in a broad spectrum of *in vitro* and *in vivo* models. In this section, we will provide an overview of this technology and describe its possible application for the characterization of cancer genes.

22.2.1 Controlled Cas9 Expression by Fusion to a Destabilized Domain

Previous studies demonstrated that the presence of an FKBP12-derived destabilizing peptide (DD-) induces the rapid degradation of the fused-protein by the proteasome system (Banaszynski *et al.*, 2006; 2008). Yet, in the presence of a synthetic ligand, Shield-1, the peptide and the fusion partner are effectively “shielded” from degradation (Banaszynski *et al.*, 2006; 2008). To this end, we developed a dual lentiviral vector system consisting of: (1) a first module, in which the sgRNA cassette is driven by the constitutive U6 promoter and (2) a second module, in which Cas9 is fused at its N-terminus with a ligand-responsive destabilizing domain derived from an engineered FKBP12 mutant (DD-Cas9) under the control of the elongation factor-1 α short (EFS) promoter (Figure 22.1A and B).

This vector was engineered based on the LentiCRISPR V2 (Addgene) backbone. The high-affinity ligand-dependent DD of a mutant FKBP12 protein (F36V, L106P) was inserted at the N-terminus of SpCas9. The DD-sequence was amplified from a degron-donor backbone (pMK backbone). The BsmBI digestion site in the DD-sequence was replaced with silent mutations via site-directed mutagenesis. All PCR-based amplifications of cloning templates were performed with Phusion High Fidelity DNA Polymerase (M0530, New England Biolabs). Primers used in cloning procedures and in sequence analysis are listed in Table 22.1. To package the Virus Hek-293T cells we transiently co-transfected on a 10 cm culture dish with 3 μ g of the construct encoding the genes of interest and sgRNA, along with 6 μ g of the packaging plasmid psPAX2 and 3 μ g of the envelope plasmid pMD2.G (Addgene) using 30 μ L of the Lipofectamine 2000 reagent (Life Technologies). 10 mL viral particles were collected after 48 hours of transfection by clarifying the supernatant through 0.45 μ m filter membrane (GE Healthcare). Virus

Table 22.1 Complementary oligonucleotides used for cloning sgRNAs and PCR primers for surveyor assay

Oligo	Sequence
CypD g.131 F	5'- CACCgCAGGTACACGAGCGGGTTCC -3'
CypD g.131 R	5'- AAACGGAACCCGCTCGTGACCTGc -3'
CypD g.150 F	5'- CACCgTCCCCTTGCGCTCCACGTCC -3'
CypD g.150 R	5'- AAACGGACGTGGACGCCAACGGGAc -3'
EGFR g.10 F	5'- CACCgGCTGCCCCGGCCGTCCCGGA -3'
EGFR g.10 R	5'- AAATCCGGGACGGCCGGGGCAGCc -3'
EGFR g.60 F	5'- CACCgTCCTCCAGAGCCCGACTCGC -3'
EGFR g.60 R	5'- AAACGCGAGTCGGGCTCTGGAGGAc -3'
RPA3 g.25 F	5'- CACCgCCGGCGTTGATGCGCGACCT -3'
RPA3 g.25 R	5'- AAACAGGTCGCGCATCAACGCCGGc -3'
RPA3 g.44 F	5'- CACCgGATGAATTGAGCTAGCATGC -3'
RPA3 g.44 R	5'- AAACGCATGCTAGCTCAATTTCATCc -3'
TP53 g.140 F	5'- CACCgCCATTGTTCAATATCGTCCG -3'
TP53 g.140 R	5'- AAACCGGACGATATTGAACAATGGc -3'
tdTOM g.230 F	5'- CACCgTTGTAATCGGGGATGTCGGC -3'
tdTOM g.230 R	5'- AAACGCCGACATCCCCGATTACAAC -3'
tdTOM g.282 F	5'- CACCgGGAGCGCGTGATGAACTTCG -3'
tdTOM g.282 R	5'- AAACCGAAGTTCATCACGCGCTCCc -3'
tdTOM g.292 F	5'- CACCgGAACTTCGAGGACGGCGGTC -3'
tdTOM g.292 R	5'- AAACGACCGCCGTCCTCGAAGTTCc -3'
EGFR-SVR-F	5'- TTGGCTCGACCTGGACATAG -3'
EGFR-SVR-R	5'- GGGGAAAGTGAGGGAAGAAA -3'
RPA3-SVR-F	5'- ATCCTGTGATCGCAGAAAGG -3'
RPA3-SVR-R	5'- CAAGACTTTGGGCCTGTTTC -3'
CYPD-SVR-F	5'- AGGGGTAGTCCACGGACAG -3'
CYPD-SVR-R	5'- CCGCACTACTGCTGGAAAC -3'
TP53-SVR-F	5'- GGGTTGGAAGTGTCTCATGC -3'
TP53-SVR-R	5'- TCAAAAGCCAAGGAATACACG -3'

transduction was optimized in order to achieve low MOI transduction. Typically, 500 µL fresh virus particles from 10 mL stock were used to infect 1x10⁶ cells on a 10 cm culture dish in 10 mL total volume of culture medium. Virus aliquots were stored at –80°C.

To test for the feasibility of this approach, we transduced our engineered lentiviral construct into A549 cells and we measured the levels of Cas9 in the presence and absence of Shield-1 by western blot and RT-PCR analyses (Figure 22.1C and D). Shield-1, obtained from Cheminpharma, was solubilized in pure ethanol and added to culture media.

An antibody against Flag was used to gauge Cas9 levels indirectly. Although Cas9 was not detected in uninfected cells or in lysates of mock-treated cells, cells treated with Shield-1 showed strong expression of the expected fusion protein in a dose-dependent manner. Importantly, all experimental conditions presented

with similar levels of expression of Cas9 mRNA in the presence or absence of synthetic ligand (Figure 22.1C).

We next assayed the destabilizing domain for the kinetics of protein degradation. Cells were treated with Shield-1 and harvested 2, 6, 12, 24, 48 and 72 hours after ligand treatment. Upon just 2 hours of treatment with Shield-1, we observed a rapid stabilization of Cas9 protein compared with uninfected and mock-control cells (Figure 22.1D). This effect was reversible upon 2 hours of ligand withdrawal from the media, with protein levels becoming negligible within 12 hours (Figure 22.1D). The fact that Cas9 was not detectable in mock-control cells or upon Shield-1 withdrawal indicates that the fusion of Cas9 with the engineered FKBP12 destabilized mutant holds Cas9 expression under tight control (Figure 22.1C and D).

Taken together, these data demonstrated that – when fused to Cas9 – FKBP12-derived destabilizing domains enable tight, rapid and reversible control of Cas9 expression, making DD-Cas9 a suitable tool for the generation of inducible genome editing systems.

22.2.2 Temporal Control of Gene Editing by Destabilized Cas9

To assess whether DD-Cas9 was competent for gene editing, we targeted multiple genes controlling diverse biological processes such as oncogenic transformation (EGFR), DNA damage responses (TP53), mitochondria metabolism (CypD) and DNA replication/repair (RPA3). A549 cells were transduced with lentiviral constructs expressing DD-Cas9 and locus-specific guide RNAs (Figure 22.2A). Cells were treated with the ligand 24 hours after lentiviral infection. Cell extracts were collected after six days and immunoblotted to verify gene editing (Figure 22.2A). The efficient formation of on-target indel mutations in cells treated with Shield-1 was confirmed via the SURVEYOR assay (Figure 22.2B) and in the case of TP53 by deep sequencing (Figure 22.2C–E).

Cells were infected with lentivirus and incubated at 37 °C for indicated times as described. Genomic DNA was extracted using mini-spin columns following the manufacturer's instructions (Macherey-Nagel). In brief, cell pellets were resuspended in lysis buffer and incubated at 56 °C for 3 hours, followed by 70 °C for 10 minutes. SpCas9-induced mutations were detected using the SURVEYOR Mutation Detection Kit (Transgenomic/IDT). An approximately 900 bp region surrounding the CRISPR mutation target site was PCR-amplified using Phusion High-Fidelity DNA Polymerase (NEB). Single band PCR products were purified using USB PrepEase Gel Extraction kit (Affymetrix). 450 ng purified PCR fragments mixed with 2 µL 10× Taq DNA Polymerase PCR buffer in a total volume of 20 µL were subjected to a series of melt and reanneal temperature cycles with gradual increments lowered in each thermal cycle. SURVEYOR nuclease and SURVEYOR enhancer were then added to the reaction mixture in order to selectively digest heteroduplex DNA substrates. Digested products were run and visualized on a 1.2% agarose gel. Band intensities were quantified using ImageJ software. Indel percentage was calculated using the formula $100 \times (1 - (1 - (b + c)/(a + b + c))^{1/2})$, where *a* stands for the integrated intensity of the undigested product and *b* and

c are the integrated intensities of each cleaved PCR product. Primers used for SURVEYOR assay are listed in Table 22.1.

Target region primers (F-CCTGGTCTCTGACTGCTCT, R-CTGCCCTGGTAG GTTTTCTG) surrounding the TP53 guide RNA (gRNA.140) recognition site were designed using the batch version of Primer3. The genomic locus was amplified with Phusion High-Fidelity DNA Polymerase (NEB) using 100 ng genomic DNA as input and 20 cycles of PCR. Amplicons were purified using Nucleospin Gel and PCR Clean-up (Macherey-Nagel). Sequencing libraries were prepared using the NEBNext Ultra DNA Library Preparation kit for Illumina (cat# E7370), strictly following the manufacturer's instructions. Briefly, 50 ng of input template was end-repaired, tagged with Illumina adaptors and treated with USER enzyme, followed by limited-cycle PCR with indexing primers. After this amplification, DNA was purified with Agencourt AMPure XP beads (Beckman Coulter). Then, barcoded libraries were pooled in equimolar ratios, quantified by a qPCR library quantification kit (KAPA Biosystems) and subjected to massively parallel 150 bp pair-end sequencing using Miseq Instrument with Miseq Reagent 300 Cycles Kit v2 (Illumina).

To understand the sequencing output, Illumina reads were processed as previously described, with minor modifications (Shi *et al.*, 2015). To summarize, paired forward and reverse Illumina reads were stitched together if they overlapped by at least 15 bp at their 3' ends and contained at least 40 bp of wild-type sequence in their 5' ends. These "observed sequences" were then aligned with the wild-type sequence starting from both ends and moving inward. Forward and reverse breakpoints were defined for each observed sequence as positions where the aligned wild-type sequence and observed sequence first began to differ by at least two consecutive mismatches. The region of wild-type sequence between these breakpoints was defined as the "deletion," and the region of observed sequence between these breakpoints was defined as the "insertion." Figure 22.2D and E illustrates a pile-up of the observed deletion sequences and provides a histogram showing the lengths of the insertion and deletion regions.

Figure 22.2E also illustrates an alignment of the 20 most frequently observed patterns near the Cas9 target sequence, as well as the counts of these patterns for each of the samples. Interestingly, the sequence alignment shows that some observed sequences had very long insertions. These long insertions usually came paired with long deletions, a phenomenon that likely represents an artifact of the breakpoint calling algorithm arising from a region of low sequence quality in the reverse reads. This artifact, however, does not affect the conclusions drawn from this data set.

Overall the data analyses indicated that induction of Cas9 with Shield-1 led to efficient gene editing, and resulted in a significant reduction in the expression of EGFR, CypD, RPA3 and p53 proteins. Notably a comparison between inducible DD-Cas9 and constitutive Cas9 also indicated similar levels of efficiency of gene editing (Figure 22.2E).

22.2.3 Functional Characterization of Cancer Genes

Next, to test whether DD-Cas9 could be used to functionally characterize genes *in vitro* and *in vivo*, we targeted the essential RPA complex. RPA is a heterotrimeric single-stranded DNA-binding complex composed of three proteins – RPA1, 2 and 3 – which stabilize replication forks by coating unwound DNA (Zou and Elledge, 2003; Zou *et al.*, 2006). The RPA complex also binds ssDNA at sites of DNA damage and recruits cell cycle checkpoint kinases (Dutta and Stillman, 1992; Zou *et al.*, 2006). Previous work showed that targeting the smallest RPA subunit, RPA3, induces degradation of all three members of the RPA complex, and is associated with decreased cell proliferation and increased cell lethality (McJunkin *et al.*, 2011; Zuber *et al.*, 2011).

A549 cells were transduced with lentiviral constructs expressing DD-Cas9 and two independent RPA3 sgRNAs (RPA3g.25 and g.44) and with Renilla sgRNA (g.208) as a control (Figure 22.3A). Cells infected with lentiviruses were plated in equal numbers in 12-well plates (BD falcon). The expression of Cas9 was stabilized using 200 nM of Shield-1 and incubated at 37 °C for indicated times, as described. Cells were rinsed twice with PBS to eliminate any floating cells, and were fixed in 4% formaldehyde in PBS (v/v) for 10–15 minutes. Fixed cells were stained by staining solution (0.1% Crystal Violet in 10% ethanol) for 20 minutes. Staining solution was aspirated from the wells, and the cells were washed with water three times in order to remove extra stain. Stained cells were air dried and imaged using the Odyssey Imaging System (LICOR). To quantify the relative cell numbers, cells were de-stained with 10% acetic acid and the absorbance of de-stained solution was measured at 590 nm at appropriate dilutions.

In the absence of Shield-1, we did not observe a difference in the number of cells when comparing uninfected cells and cells infected with lentiviruses targeting RPA3 or Renilla (Figure 22.3A). Yet, upon treatment with Shield-1, a drastic decrease in cell number was apparent in cells infected with the lentivirus harboring the RPA3, but not the Renilla guide (Figure 22.3A). Of note, analysis of the onset of this effect indicated that just 48 hours of exposure to Shield-1 compound was sufficient to induce robust decreases in cell viability.

We previously showed that the non-small-cell lung cancer (NSCLC)-derived PC9 cells harbor an oncogenic-EGFR mutation and are highly dependent on EGFR for their survival, unlike EGFR wild-type cells (i.e. A549 cells; Yao *et al.*, 2010). We took advantage of this observation to test whether DD-Cas9 could be used for the identification of genotype-driven vulnerabilities. PC9 cells were transduced with lentiviral constructs carrying the guide RNAs targeting the EGFR locus (EGFR g.10 and g.60) (Figure 22.3B). As demonstrated with Crystal Violet staining and cell counting, treatment of PC9 cells with Shield-1 resulted in a prominent and rapid decrease of viable cells (Figure 22.3C and D). This was not observed in A549 cells transduced with constructs targeting Renilla or the EGFR locus; however, western blot analysis of cell extracts confirmed decreased expression of EGFR in these cells.

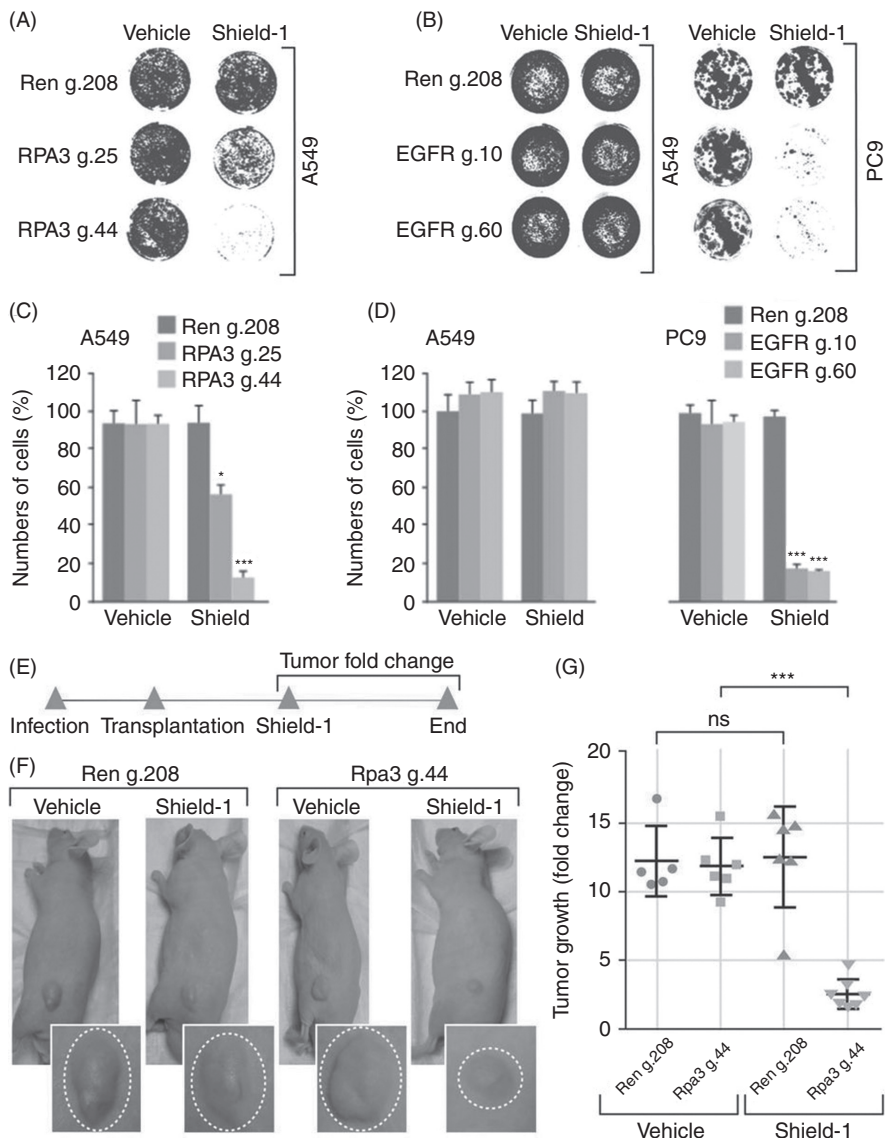


Figure 22.3 DD-Cas9 can drive robust synthetic ligand-dependent gene editing in mammalian cells and enables the identification of essential genes *in vitro* and *in vivo*. Targeting RPA3 in A549 cells and EGFR in PC9 cells resulted in a dramatic decrease in the number of viable cells following induction with Shield-1. (A) A549 cells were transduced with the DD-Cas9 vector targeting the RPA3 locus and, as a control, the Renilla gene. (B) PC9 cells and A549 cells were transduced with DD-Cas9 vector targeting the EGFR locus. In all cases cells were treated 24 hours after infection with Shield-1 (200 nM) for six days. The number of cells was assessed after six days by Crystal Violet staining. (C, D) The data in the charts represent quantification of Panels A and B, respectively, and are expressed as mean $\pm$ standard deviation upon treatment with Shield-1 and vehicle control relative to uninfected cells ($n = 4$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$) Student's t-test. (E) Schematic of the transplantable model system used in this study. (F) A549 cells were transduced with DD-Cas9 targeting RPA3 and as control Renilla and transplanted sub-cutaneously in immune-deficient mice. When the tumors reached an approximate size of 4–5 mm in diameter, mice were treated with Shield-1 (1 μ g). Fold change in tumor growth was determined ten days after treatment. Pictures illustrate representative tumor sizes. The chart in panel (G) illustrates combined quantification of tumor volume in the indicated cohorts upon Shield-1 treatment. Data represent mean $\pm$ standard deviation and are shown relative to the negative control (Cas9 negative) *** $p < 0.0001$ (RPA3 g.44 + Shield-1 vs. – Shield-1) Student's t-test (Senturk *et al.*, 2017). (A black-and-white version of this figure will appear in some formats. For the color version, please refer to the plate section.)

The Shield-1 compound has been shown to be non-toxic and to display favorable pharmacodynamics and kinetic properties in mice models (Banaszynski *et al.*, 2006). Having shown ligand-dependent gene knockout in *in vitro* cell culture, we reasoned that this platform could be exploited for assessing gene function *in vivo* as well. As a proof of principle, we transplanted A549 cells transduced with the DD-Cas9 vector targeting either RPA3 or Renilla into immune-compromised mice via subcutaneous injection (Figure 22.3E–G). When tumors reached an approximate size of 4–5 mm in diameter, mice were treated with Shield-1 or vehicle for four days and tumor size was assessed for the next ten days. We observed a drastic decrease in tumor growth in tumors expressing RPA3 sgRNA upon Shield-1 treatment (Figure 22.3F and G). To confirm the decreased expression of RPA3, we used immunostaining (not shown).

22.2.4 Design of a Traceable System Based on Destabilized Cas9

An advantage of the described system is the conditional regulation of Cas9 protein stability independent of its mRNA expression. This implies that any gene of interest can be co-expressed with destabilized Cas9 from the same promoter. Among various strategies employed to construct bi-cistronic or multicistronic vectors, an internal ribosomal entry site (IRES) has been widely used. More recently, a self-cleaving 2A peptide has been shown to be an efficacious alternative. To verify the possibility of expressing a gene of interest together with conditionally destabilized Cas9, we cloned the fluorescent protein Venus – located downstream of DD-Cas9 – and separated them with the self-cleaving 2A peptide (P2A) (Figure 22.4A). To achieve this, we replaced P2A-linked puromycin with P2A-linked Venus – a modified version of green fluorescent protein (GFP) – for easy visualization of viral transduction efficiency as well as competition assays. To do so, P2A-Puro-WPRE-3′LTR cassette on the LentiCRISPR V2 was excised with BamHI/PmeI restriction enzymes, then a PCR product containing only WPRE-3′LTR piece was religated, which was again surrounded by BamHI/PmeI. Following this step, a PCR product containing the P2A-Venus cassette surrounded by BamHI digestion sites on both ends was amplified and cloned into the BamHI site on the plasmid in the correct orientation.

As shown in Figure 22.4B, expression of the modified-GFP fluorescent protein (Venus) was constitutively expressed, despite fusion to DD-Cas9, and was not altered by Shield-1 treatment. Similarly, the mRNA expression was found to be independent of ligand treatment (Figure 22.4B).

An important consideration when developing therapeutic strategies based on decreased expression of a gene by viral delivery is the potential impact on the physiology and viability of the transduced cells. This effect can be particularly relevant in longer time-course experiments, where subtle differences in cell growth rates can have a major impact.

In this regard, “competition assays” have been shown to be powerful systems to assess subtle cell growth changes (Zuber *et al.*, 2011). These assays are based on the competitive cell growth between transduced and non-transduced cells. The mixed culture obtained after gene transduction can be observed in the ratio of

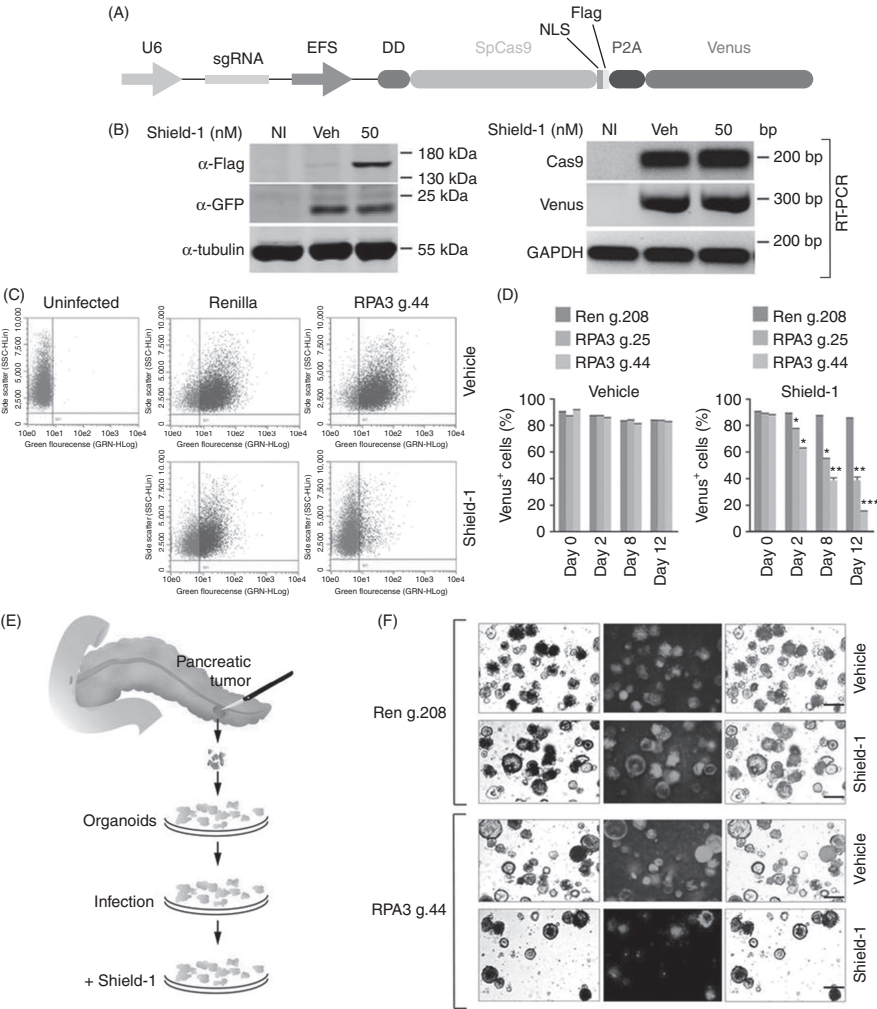


Figure 22.4 DD-Cas9 can be coupled to a modified fluorescent protein to generate a traceable system. (A) Schematic of the lentiviral DD-Cas9 plasmid containing U6 promoter-driven single-guide RNA (sgRNA) and EFS promoter-driven DD-Cas9 pieces. Note, a FLAG tag is present at the Cas9 C-terminal moiety, and 2A self-cleaving peptide (P2A) separates both DD-Cas9 and the gene of interest, in this case the modified fluorescent protein Venus. (B) Shield-1-independent expression of Venus. A549 cells were transduced with the DD-Cas9/P2A/Venus lentiviral vector and treated with Shield-1 for three days. Expression of Cas9 and Venus were assayed by western blot and RT-PCR. (C) Competition assay. A549 cells were transduced at low MOI with the DD-Cas9/P2A/Venus lentiviral vector. Cells were then treated with Shield-1 or vehicle control for 12 days and quantified by FACS. The panels represent typical FACS dot plots from A549 sorted cells. (D) A chart illustrating the percentage of Venus⁺ cells during time in A549 cells transduced with the DD-Cas9/P2A/Venus lentiviral vector targeting the indicated genes upon treatment with Shield-1 and vehicle control, error bars are standard deviation (n = 3, * p < 0.05, ** p < 0.01, *** p < 0.005) Student's t-test. (E) Schematic representing our test for cell vulnerabilities in pancreatic tumor organoids. (F) DD-Cas9/P2A/Venus could be used to infect primary human tumor-derived cultures and to assess their cellular vulnerabilities. Human pancreatic cancer-derived organoids were infected with the DD-Cas9/P2A/Venus lentiviral vector targeting RPA3 and as control Renilla. Organoids, cultured in 24-well plates, were treated with Shield-1 or vehicle control 72 hours post-infection. The number of Venus⁺ organoids was determined 72 hours after infection and six days after Shield-1 treatment (Senturk *et al.*, 2017). (A black-and-white version of this figure will appear in some formats. For the color version, please refer to the plate section.)

Venus⁺ and Venus[−] cells over time to identify minor cell growth defects. When compared to established methods of counting cells and viability assays (e.g. MTT and ATP bioluminescence), competition assays were shown to have better sensitivity.

To verify that our platform could also be used in a competition assay set-up, we infected A549 cells with the DD-Cas9/P2A/Venus lentiviral vectors targeting either RPA3 or, as a negative control, the Renilla gene at low MOI. This ensured the generation of Venus⁺ and Venus[−] mixed culture (Figure 22.4C). By tracking the ratio of Venus⁺ and Venus[−] cells infected with the RPA3-targeting vector on a time-course of 12 days, we observed a Shield-1-dependent depletion of Venus⁺ cells (Figure 22.4C and D).

The recently developed 3D “organoid” culture system allows for cultivation of both normal and patient-derived primary tumors *in vitro* (Boj *et al.*, 2015). We tested whether the DD-Cas9/P2A/Venus platform could also be applicable to this culture system (Figure 22.4E and F). We followed methods to establish and propagate human organoid cultures that were previously described by the Tuveson and Clevers laboratories (Boj *et al.*, 2015).

In brief, organoids were grown in 24-well culture plates for two days before infection. One well per infection was used. On the day of infection, organoids were dissociated into small fragments by first triturating them in media through a fire-polished glass pipette, and then by a 5-minute enzymatic digestion with TrypLE (Life Technologies) at 37 °C. The resulting small cell clusters were then resuspended with low MOI lentivirus and spinoculated at 600 RCF for 1 h at room temperature.

Upon infection, we observed approximately 80% Venus⁺ organoids. Strikingly, in organoids derived from two different tumors, after treatment with Shield-1 this fraction was decreased to less than 20% for the RPA3-targeting sgRNA (Figure 22.4F). This result clearly demonstrates the utility and efficiency of the DD-Cas9 system as a screening tool in primary tumor cells. Despite these promising results, further experiments (i.e. independent sgRNAs) will be required to validate RPA3 as a gene that is essential for primary pancreatic cells growth and/or survival.

22.2.5 DD-Cas9 Linked to Cre Allows Genetic Interaction Studies

Recent studies indicated that a CRISPR/Cas9-based approach could be used for the functional investigation of candidate genes in well-established mouse models of cancer (Maddalo *et al.*, 2014; Xue *et al.*, 2014). In particular, these studies showed that Cre-dependent somatic activation of oncogenic Kras (G12D) could be combined with CRISPR/Cas9-mediated genome editing of tumor-suppressor genes by using a lentiviral-based system that delivers both the CRISPR/Cas9 system and the Cre recombinase.

We reasoned that by generating a bi-cistronic vector for the expression of destabilized Cas9 and tamoxifen-inducible Cre, we could uncouple gene editing and Cre-mediated recombination. By providing temporal control of gene editing and activation/inactivation of LOX alleles, this approach, in

principle, could bolster precision in the study of functional interactions among genes in a variety of pre-existing mouse models derived from the Cre-Lox system. Especially in the context of cancer models, it could be used to dissect more precisely the role of mutations identified in onco-genomic studies during tumor onset and progression. Given that cancer is a multi-step process that involves the progressive accumulation of mutations, this approach could be invaluable for the identification of tumor cell vulnerabilities during the evolution of tumors from preneoplastic to fully malignant diseases.

To achieve this, we engineered the DD-Cas9/P2A/Venus backbone to express Cre-ER^{T2} downstream of Cas9 (Figure 22.5A). More specifically, we inserted an internal ribosome entry site (IRES) downstream of the DD-Cas9 open reading frame (ORF) which was then coupled to tamoxifen-inducible Cre (Cre-ER^{T2}). In this context, gene editing could be induced independently of Cre-mediated recombination. The Cre-ER^{T2} backbone was generated by introducing a stop codon at the end of the Cas9 ORF, and cloning IRES and Cre-ER^{T2} ORFs downstream of the Cas9 cassette with NLS and a Flag tag (Figure 22.5A). The Cre-ER^{T2} protein codon optimized for human expression with three naturally existing BsmBI restriction enzyme sites removed by silent mutations was synthesized, cloned into a pUC57 cloning vector and sequence-verified by Genscript. This vector served as a template for subsequent cloning experiments. IRES sequence was PCR-amplified from a TTI-GFP.

To test this system, we used genetically engineered mouse lines that express the fluorescent reporter protein tdTomato in a Cre-dependent manner from the constitutive CAG promoter (tdTom^{LSL}; Ai14). In this setting, following a Cre-mediated deletion of the floxed stop cassette, tdTomato can be readily expressed. We derived mouse embryonic fibroblasts (MEFs) and transduced them with our lentiviral vector. Three days after infection, MEFs were acutely treated with 4-hydroxytamoxifen to induce activation of Cre (Figure 22.5B). We assessed expression of tdTomato by fluorescent microscopy and FACS. TdTomato positive cells were readily observed only in MEFs infected with DD-Cas9/IRES/Cre-ER^{T2} upon 4-hydroxytamoxifen treatment (Figure 22.5C).

Next, to assess the gene editing efficiency of our system, we targeted the Cre-activated tdTomato knockin reporter allele with lentiviruses expressing sgRNAs against tdTomato (sgTom) or Renilla as a control. When we quantified sgRNA-transduced tdTomato positive MEF cells over time, we observed a substantial reduction in tdTomato expression only in those cells treated with Shield-1 compound (Figure 22.5D). Thus, this result suggests that the unique system that we have developed is a functional tool for the gene editing of an endogenous allele, and that Cre and Cas9 activity may be successfully uncoupled.

Consummately, these results demonstrate the potential of using destabilizing domain technology to generate scalable and efficient platforms for the temporal control of CRISPR/Cas9-based gene editing. Moreover, the described method enables the testing of co-operativity between genetic events and the identification of tumor cell vulnerabilities in both *in vitro* and *in vivo* models.

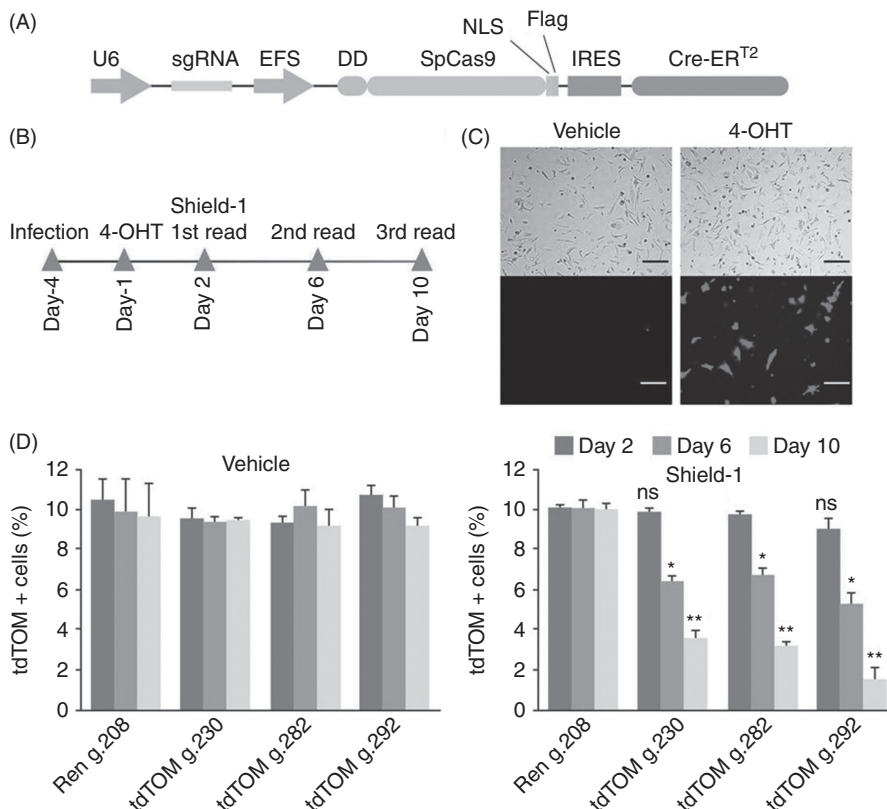


Figure 22.5 DD-Cas9 can be combined with tamoxifen-inducible Cre (Cre-ER^{T2}) to facilitate the study of genetic interactions *in vivo* in pre-existing mouse models based on the Cre-lox system. (A) Diagram of the DD-Cas9/IRES/Cre-ER^{T2} vector utilized in this study. (B) Schematic for experimental design. MEF cells obtained from CAG-LSL-tdTomato mice were transduced with the indicated vectors. Induction of Cre-mediated recombination was induced upon 4-hydroxytamoxifen treatment (5 μM) for three days. To test for efficiency of gene inactivation, cells were treated with Shield-1 (200 nM). (C) Bright-field and fluorescence images of LSL-tdTomato cells treated with 4-hydroxytamoxifen. (D) The chart represents the percentage of tdTomato-positive MEFs transduced with DD-Cas9/IRES/Cre-ER^{T2} lentiviruses carrying the indicated sgRNAs over time after exposure to Shield-1 and vehicle. Numbers of tdTomato-positive and -negative cells were determined by FACS. Each bar represents the average of three individual replicates, error bars are standard deviation (n = 3, *p < 0.05, **p < 0.01) Student's t-test (Senturk *et al.*, 2017). (A black-and-white version of this figure will appear in some formats. For the color version, please refer to the plate section.)

22.3 Discussion

The possibility of acutely inactivating genes in cells and in organisms is essential for our understanding of gene function. Lethality is the most conclusive phenotype defining a gene as “essential.” In the context of cancer research, studying essential genes is particularly important as they could serve as potential therapeutic targets. Yet, the fact that their inactivation leads to cell death renders them the most difficult class of genes to study.

CRISPR/Cas9 technology has been widely utilized to create heritable changes in the genome (Doudna and Charpentier, 2014). However, the inactivation of genes often results in cell lethality, functional deficits and developmental defects, limiting the utility of such approaches for studying tumor gene function. Additionally, constitutive expression of Cas9 may result in toxicity and in the generation of off-target effects. To overcome these limitations, we generated a temporally controlled Cas9 expression system by fusing Cas9 to an engineered mutant of the human FKBP12 protein (DD-Cas9).

22.3.1 The Choice of an Inducible Vector

Many different approaches have been developed and used in the past for the identification and characterization of essential genes. These methods include: (1) the generation of temperature-sensitive mutants, (2) the generation of conditional deletion alleles in which a gene of interest is flanked by loxP recombination sites to enable acute gene inactivation upon expression of Cre recombinase and (3) inducible gene knockdown based on shRNA-mediated silencing (Premsrirut *et al.*, 2011; Tyner *et al.*, 2002; Ventura *et al.*, 2007). More recently, methods to temporally control CRISPR/Cas9-based genome editing technologies have also been developed. These systems are based either on the transcriptional control of Cas9 and/or sgRNA or on Cas9 post-transcriptional/translational activation (Davis *et al.*, 2015; Dow *et al.*, 2015; Wright *et al.*, 2015). Regarding the former, multiple methods have been developed based on the use of DOX-induced sgRNA and Cas9 expression. In the latter case, Cas9 activity is regulated either by generating a split Cas9 enzyme that could be restored upon treatment with rapamycin (a light-inducible Cas9) or by inserting a 4-HT-intein in the Cas9 transcript that could be spliced by treatment with tamoxifen. As an alternative to these existing CRISPR/Cas9 techniques, we utilized a conditionally destabilized Cas9. By exploiting a strategy developed by the Wandless lab, we showed that by conjugating a genetically encoded destabilizing domain to Cas9 we could achieve a highly specific conditional regulation of Cas9 expression and gene editing (Senturk *et al.*, 2017). Compared to other methods, this approach enables the generation of a system that would: (1) provide potent, robust and temporally controlled gene editing, (2) be applicable to a broad spectrum of cell types and tissues, (3) facilitate high-throughput manipulation and (4) be traceable.

As the small-molecule DD ligand Shield-1 is highly permeable and non-toxic in cultured cells and animals, we also demonstrated our methods could be utilized in transplantable model systems and in mosaic mouse models (Senturk *et al.*, 2017). In particular, by coupling DD-Cas9 with tamoxifen-inducible Cre, we showed that our lentiviral vector could, in principle, facilitate the study of genetic interactions *in vivo* in a variety of pre-existing mouse models of human diseases based on the Cre-Lox system.

Recent studies have shown that a similar lentiviral vector based on tandem expression of Cas9 and Cre recombinase could be used *in vivo* to rapidly evaluate human cancer genome candidates (Senturk *et al.*, 2017). Our approach enables the temporal and independent control of gene editing and Cre-mediated

recombination. Therefore, in principle, this method uniquely informs the investigation of the roles of putative cancer genes in tumorigenesis and in driving tumor vulnerabilities.

Although we have described here the use of CRISPR/Cas9 technology for the characterization of cancer genes, this same approach could be easily implemented for the identification of genes that are required for the survival or progression of tumors.

22.3.2 Future Development

CRISPR/Cas9-based gene editing is an ever-evolving field, and novel Cas9 mutants with different properties are constantly being generating. The efficiency and simplicity of the Shield-1 degron Cas9 approach facilitates the easy adaptation of this system to explore novel research objectives. The amenability of this system has been demonstrated, for example, by the design and use of catalytically inactive Cas9 to activate or repress transcription (Chavez *et al.*, 2015).

In addition to the Shield-1-based system, many other destabilizing domains have been developed. In principle, these other approaches can be adapted for the generation of novel inducible Cas9 systems. In preliminary experiments, we have shown that the *E. coli* dihydrofolate reductase (ecDHFR) could be fused to Cas9 to destabilize it (Iwamoto *et al.*, 2010). Similar to our observations when we treated cells with Shield-1, our data suggest that cells treated with the small-molecule ecDHFR ligand trimethoprim (TMP) stabilized the ecDHFR-Cas9 protein in a rapid, reversible and dose-dependent manner. Compared to Shield-1, TMP is inexpensive and capable of more efficiently traversing the blood–brain barrier; therefore, its use will enable the regulation of gene editing in the mammalian central nervous system (Iwamoto *et al.*, 2010).

22.4 Concluding Remarks

Cancer is a multi-step process characterized by the accumulation of multiple genetic aberrations over time. The development of CRISPR/Cas9 technologies has both simplified and enhanced the efficiency of gene editing approaches. By rapidly introducing mutations in cell lines, specific tissues and organisms, the CRISPR/Cas9 system has enabled whole-genome and targeted loss-of-function and gain-of-function cancer screening. The implementation of CRISPR/Cas9 systems has already facilitated the identification and characterization of multiple novel oncogenes, tumor-suppressor genes and cancer non-coding regulatory regions. CRISPR/Cas9 approaches have also allowed the development of animal models of cancer at an unprecedented pace. The generation of conditionally destabilized Cas9 could prove to be an important tool enabling the temporal control of gene editing. This novel platform is robust, easily scalable and, therefore, suitable for the identification and/or characterization of essential genes and the interrogation of the functional interaction of genes *in vitro* and *in vivo*.

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